

1. Quantum-Inspired Hybrid Genetic Algorithm and Graph Neural Network Ensemble for Multimodal Classification

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Abstract: High-dimensional and heterogeneous datasets pose significant challenges for classification due to complex feature interactions and varied data structures. This paper proposes a quantum-inspired hybrid genetic algorithm (QIHGA) combined with a graph neural network (GNN) to address these challenges within a multimodal ensemble framework. QIHGA introduces qubit-based representation and superposition mechanisms to enhance global search efficiency, improving feature selection and model parameter optimization in large feature spaces. In parallel, the GNN module constructs graph representations of data to capture relationships among features or samples, enabling effective learning of structured embeddings. The ensemble integrates optimized feature subsets and GNN-derived embeddings through a fusion layer to produce final classification outputs. This design leverages complementary strengths: QIHGA enhances search capability and identifies informative features, while the GNN models interaction patterns that traditional methods often overlook. The combined model improves accuracy, robustness, and interpretability, as feature selection highlights influential variables and graph structures reveal relational importance. Experimental results show that the proposed hybrid model outperforms traditional machine learning classifiers, achieving higher accuracy and robustness in complex multimodal classification tasks. This approach offers a promising direction for addressing complex classification tasks involving multimodal and relational data. © 2025 IEEE.

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